

ColloseumTwoChoiceCounterRun1

February 27, 2025

1 Counting two-choice (or valence) experiment

1.1 Notes

Overview:

Rationale:

Date of Experiment:

Genotypes:

Sex:

Light Paradigm:

Design:

Misc:

1.2 Configuration

Configuration was standard. No new or untested code.

```
[22]: #!/autoreload 1  
import warnings  
from Arena import *  
import Tracker  
import TwoChoiceTracker  
from ExperimentalDesign import *  
from Parameters import *  
  
# Suppress all warnings  
warnings.filterwarnings('ignore')
```

1.3 Parameters

Just a standard arena max, accepting the default values and a Two-Choice Tracker.

```
[23]: p=Parameters()  
      #p.set_small_arena_values(TrackingType.TWOCHOICECOUNTER)  
      #p.set_obscura_values(TrackingType.TWOCHOICECOUNTER)  
      p.set_colloseum_values(TrackingType.TWOCHOICECOUNTER)  
      p.print()
```

```
tracking_type: TrackingType.TWOCHOICECOUNTER  
fps: 0
```

```
mm_per_pixel: 0.145
speed_window_seconds: 1
micromove_speed_mm_sec: [0.2, 2]
walking_speed_mm_sec: 2
sleep_threshold_min: 5
interaction_distances: [8]
```

1.4 Read in the data

Next we will read in the data to an Arena object. The format is

```
Arena(parameters, ExperimentName, data_path=".").
```

This will automatically calculate all relevant metrics for the tracking type in question.

The predefined parameters object is required. *ExperimentName* and *data_path* are optional. if no data directory is specified, it defaults to the current working directory and if no experiment name is provided it will use the first .xlsx file in the data directory.

The following files must be present in the defined data directory : * The DTrack experiment file, which is an excel sheet with the name '*ExperimentName.xlsx*'. * The DTrack data files themselves, which are named '*ExperimentName_Data_1.csv*', '*ExperimentName_Data_2.csv*', etc. * An experiment file, which is named '*ExperimentName_Design.txt*'. Note that you can incorporate flipped axes to orient lights for data visualization.

```
[24]: arena = Arena.Arena(p,data_path='./Data/')
```

1.5 Summary

Get the summary results for all trackers, including treatments, etc.

This will copy the results to the clipboard and save the data in a file called '*ExperimentName_Summary.csv*'.

The semicolon at the end of the command will suppress the output to this notebook. Remove it if you want to see the results dataframe.

```
[25]: arena.summarize(copy_to_clipboard=True,write_to_csvfile=True);
```

Or you can also create a faceted summary if you want to separate the different phases of the experiment.

The file name will be '*ExperimentName_Summary_Facet.csv*'.

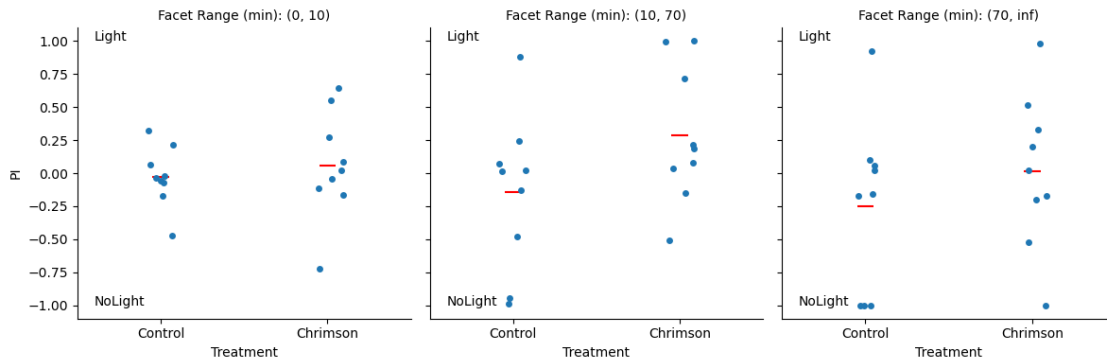
```
[26]: arena.
      ↪ summarize_facet(cutoffs=(10,70),copy_to_clipboard=True,write_to_csvfile=True);
      ↪
```

1.6 Plotting

There are lots of cool plotting funtions. Some are shown below.

Facet plots: First of PI, then Percentage. Distance metrics don't apply to counting experiments. Cutoffs separate the acclimaton phase from the experiment phase from the cool down phase.

```
[27]: arena.plot_pi_facet(cutoffs=(10,70))
```



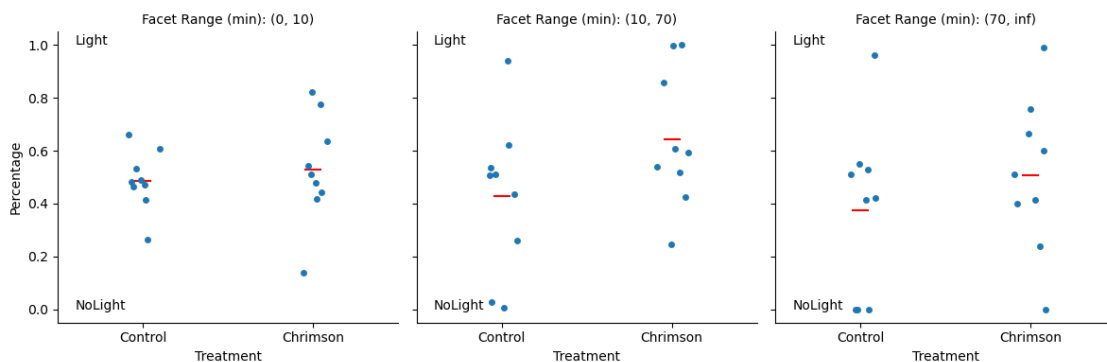
```
[28]: arena.run_pairwise_comparisons_facet(metric="FinalPI",cutoffs=(10,70))
```

```
##### T-Test #####
Column = FinalPI, Range Minutes = (0.00 , 10.00)
Control vs. Chrimson: T=-0.54, p=0.59630

##### T-Test #####
Column = FinalPI, Range Minutes = (10.00 , 70.00)
Control vs. Chrimson: T=-1.65, p=0.11892

##### T-Test #####
Column = FinalPI, Range Minutes = (70.00 , inf)
Control vs. Chrimson: T=-0.91, p=0.37520
```

```
[29]: arena.plot_percentage_facet(cutoffs=(10,70))
```



```
[30]: arena.run_pairwise_comparisons_facet(metric="FinalPercentage",cutoffs=(10,70))
```

```
##### T-Test #####
```

```
Column = FinalPercentage, Range Minutes = (0.00 , 10.00)
```

```
Control vs. Chromson: T=-0.54, p=0.59630
```

```
##### T-Test #####
```

```
Column = FinalPercentage, Range Minutes = (10.00 , 70.00)
```

```
Control vs. Chromson: T=-1.65, p=0.11892
```

```
##### T-Test #####
```

```
Column = FinalPercentage, Range Minutes = (70.00 , inf)
```

```
Control vs. Chromson: T=-0.91, p=0.37520
```

```
[31]: arena.plot_trackers_pis()
```

